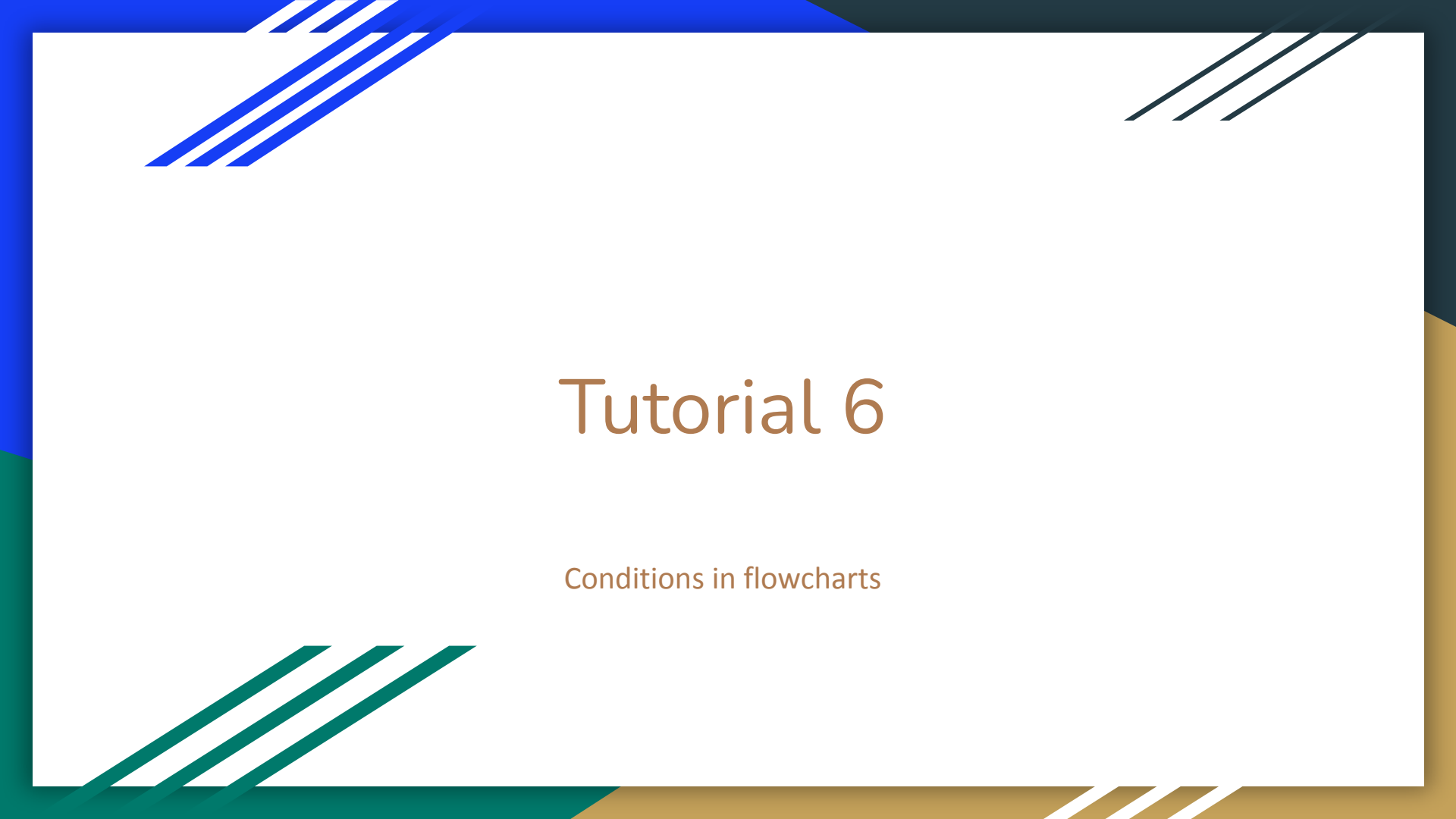
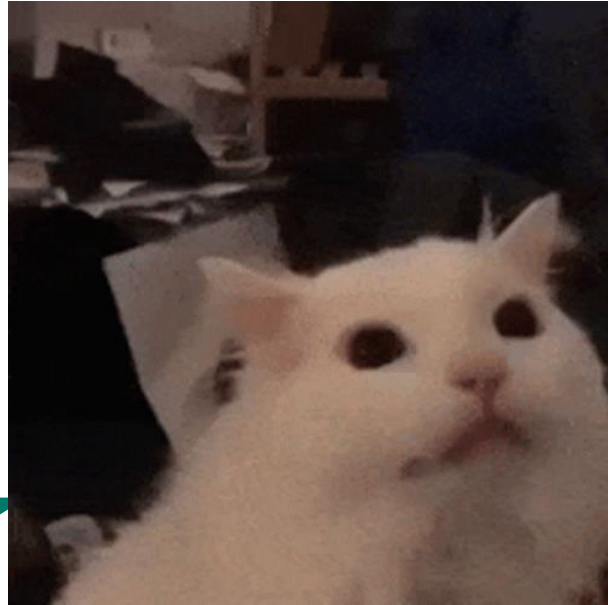




Tutorial 6

Conditions in flowcharts

How was the test?



Overview

Revision

- Flowchart
- Comparison operators
- Logical operators

Conditional structures

- Conditions in flowcharts
- Ternary-if statements
- Nested ternary-if

Revision: Flowchart

- Flowcharts provide a visual abstraction of how parts/patterns interact.
 - makes it easier to translate and develop the algorithm in a programming language.
- ISO 5807 provides a list of common flowchart symbols:
 - Start/End, Processing, Predefined process, Decision, Input/Output, Flow arrows, Annotations or (notes), Connectors
- Flowchart structures are high-level patterns to illustrate:
 - linear execution, i.e. sequence structures
 - decision making, i.e. selection structures
 - repeating until a condition is met, i.e. repetition structures

Revision: Comparison operators

- comparison operators are used to compare the values of two operands and determine their relationship.
- Operators return a boolean value (true or false) based on whether the comparison is true or false.
 - Equal to (==)
 - Not equal to (!=)
 - Greater than (>)
 - Less than (<)
 - Greater than or equal to (>=)
 - Less than or equal to (<=)
- comparison operators are commonly used in conditional statements,
 - such as if statements and loops,
 - to control the flow of program execution based on the evaluation of conditions.

Revision: Boolean expression

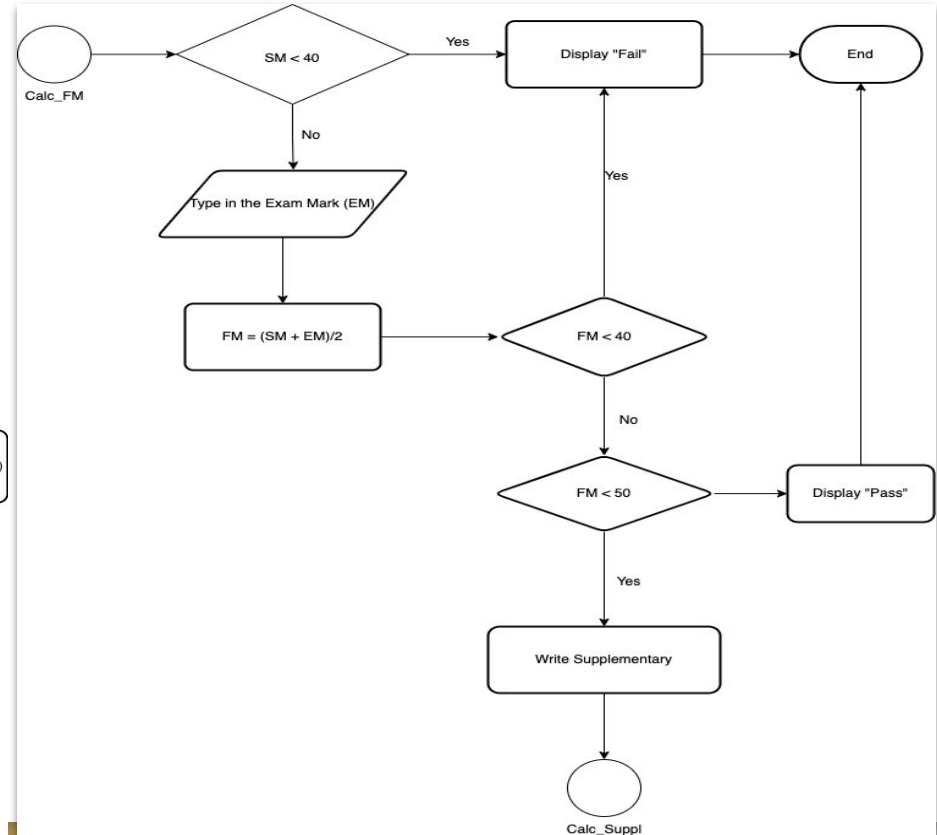
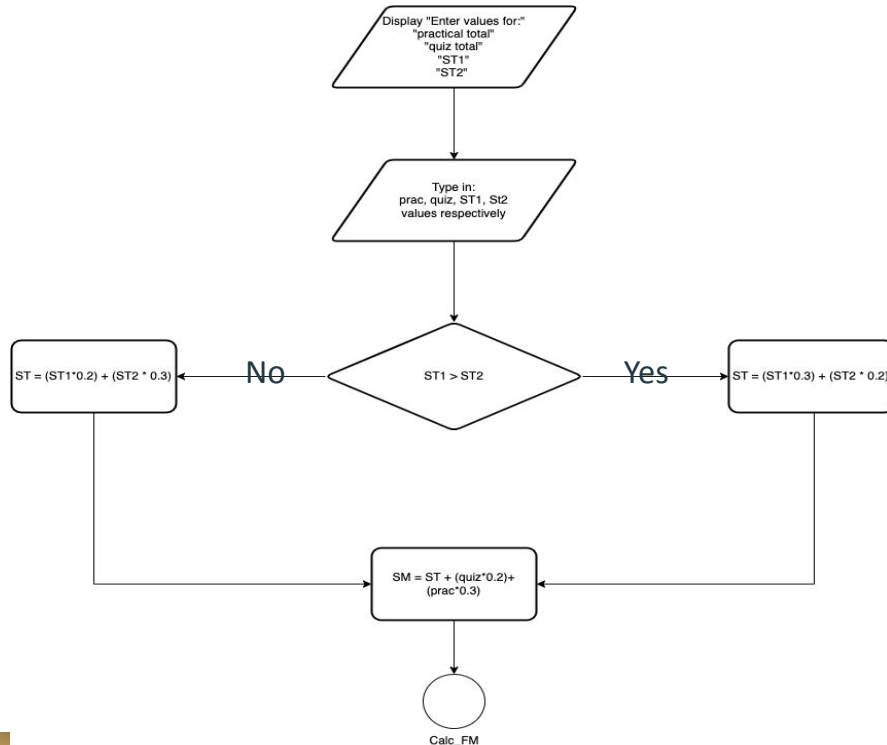
- Expressions that either results in a true or false.
 - eg. comparison operators, $a > b$, will always return either true or false
- Expressions can consists of a single or multiple condition(s)
 - Conditions that comprise more than one expression are joined by the binary logical (Boolean) operators `||` (or) and `&&` (and)
 - Example:
 - Single condition: $a \leq b$
 - Multiple condition: $a \leq b \ \&\& \ b > c$

Conditional structures: Introduction

- The selection statement only allows specific commands to be executed depending on the outcome of the condition.
- There are three selection statements defined in C++:
 - the Ternary-IF statement,
 - IF-statement
 - the SWITCH-statement.

Conditional structures: Conditions in flowchart

Identify selection statements



Conditional structures: Ternary-if statement

- Also known as the conditional operator
- It takes three operands:
 - a condition, a result for true, and a result for false.
- Syntax:
 - **<condition>** ? **<expression_true>** : **<expression_false>**;
 - **<condition>**: Must be a boolean expression
 - **<expression true>** and **<expression false>**: Any expressions that result in a value
 - Literals (42, 3.14, 'c', etc.)
 - Variables (myVar, result, etc.)
 - Function calls (calculate(), getMax(a, b), etc.)
 - Arithmetic expressions (a + b, a * b, etc.)
 - Both the **<expression true>** and **<expression false>** need to be of the same type, or convertible to the same type.
- Examples: (online compiler)

Conditional structures: Nested ternary-if

- Ternary Operators can be nested arbitrarily many times,
- as long as in the end each expression (and sub-expression) evaluates to the same (or implicitly convertible) types.
- Example:

```
String mark = ( score >= 90 ) ? "High Distinction " :  
              ( score >= 75 ) ? " Distinction " :  
              ( score >= 60 ) ? " Credit " :  
              ( score >= 50 ) ? "Pass " : "Fail " ;
```

- Disadvantage: challenging to read and maintain.
 - It's best used for simple conditions and when clarity is not compromised.

End